

Plant Station R17 - Test and Acceptance Sheets

Controlled field	Value
Document number	PLANT-TS-001
Plant design revision	R17
Document issue	D02
Issue date	28 September 2026
Preceding document	New test-sheet family; no preceding TS issue located
Status	Issued for prototype review; physical acceptance open

Current mechanical configuration: R17 integrated body, R16 lid (R15 geometry), R13 fascia and planter. Electrical reference: frozen Uno UI v6 source and 27 September interface record. Fit, powered wet operation and unattended operation remain unaccepted.

R17 incorporates R15/R16 mechanical changes and the current Uno UI v6 interface. Previous issues are retained. Blank physical results remain open; no acceptance signature is inferred.

Revision control

Narrow parallel left-margin bars identify surviving changes against the verified matrix. Dates are documentary dates. R15/R16 source changes are incorporated without creating intervening manual issues.

Revision / class	Change and affected content
R5-R13 / source history	Earlier assembly and mechanical changes remain traceable through the preserved D01 matrix. Missing document editions are not invented.
TS family / D02	PLANT-TS-001 assigned for this test-sheet set. Prior tests are source evidence; no earlier same-family controlled test-sheet issue is asserted.
R15 / 27 Sep 2026	Reinforced Uno bosses, thicker recessed-head lid and additional pump relief; source changes incorporated at R17, physical trials open.
R16 / 27 Sep 2026	Four relay bosses below Uno; source change incorporated at R17; actual underside and hole pitch unmeasured.
UI v6 / 27 Sep 2026	Three buttons and integrated lamps, enabled active-HIGH relay, 1-4 s request, soil display and 60 s backlight timeout. Firmware revision is independent of mechanical R17.
R17 / 28 Sep 2026	Mounted USB-C cut and pilots, adjacent diameter 10 DC extension hole; estimated dimensions authorised; coupon fit required.
D02 / 28 Sep 2026	Integrated current R17 document issue. Uses the established enhanced template, controls and cumulative revision bars. No R15/R16 manual issues created.

D02 / document control: new integrated R17 technical content in the established D01 A4 template. Pagination, new headings and editorial reflow alone receive no technical bar.

R5 is the earliest located assembly guide. R1-R4 and R9-R12 PDFs were not located in the previous inventory. R15/R16 are evidenced mechanical prototype changes, incorporated here without fictitious intermediate PDF/manual editions. Margin labels identify documentary changes, not physical test passes.

List of effective pages

All pages are effective at D02, dated 28 September 2026, for the R17 prototype. Section references provide navigation; technical change labels are controlled separately.

Page	Issue	Content
1	D02	Control pages
2	D02	Control pages
3	D02	Control pages
4	D02	Section 1
5	D02	Section 2
6	D02	Section 3
7	D02	Section 4
8	D02	Section 5
9	D02	Section 6

Section / page cross-reference

Section	Subject	Effective pages
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R17

01 Test article, prerequisites and recording rules

Test article: one R17 prototype using the controlled selected print files and frozen Uno UI v6 source. This sheet set defines future physical tests; it records no new physical passes. Link each action to a build ID, source hash, operator/date, actual configuration and retained evidence.

Record field	Actual value
Build ID / serial / operator / date	_____
Printed-part hashes / document issue	_____
Printer / material / slicer / orientation	_____
Component models/revisions and photographs	_____
Screw measurements / engagement	_____
Supply identifiers / ratings / protections	_____
Instruments and calibration/check status	_____
Test method / environmental conditions	_____
Deviation IDs / defect report / evidence location	_____

CAUTION

Before electrical work disconnect every supply. Before powered logic/upload checks confirm the separate pump supply is OFF. No upload, printer operation or powered test is executed by this document.

Use Pass / Fail / Not tested / Deferred with evidence. A blank actual result remains Not tested. Acceptance tolerances not supplied must be agreed and recorded before assessing that test; do not invent acceptable duration, volume, torque or leak rates.

02 Mechanical coupons and dry assembly

Action ID	Test	Expected acceptance	Existing status
A17-01	USB/DC fit coupon	USB shell and screw pattern fit; plug engages; DC thread/nut span 6 mm; no cracks	Not tested
A17-02	Uno screws and board	2.8 pilot trial takes actual screws; all four centres register without PCB bowing	Not tested
A17-03	Flush lid	Heads sit flush/below surface; pockets intact; lid seats; engagement verified	Not tested
A17-04	Pump route	Actual pump/tube clear relief; no kink; dry partition preserved	Not tested
A17-05	Relay fit	Four-hole registration; solder, screws, terminals and service removal clear	Not tested
A17-06	Fascia and full dry build	LCD/buttons/pot fit; wiring untrapped; planter lifts clear; all fasteners retain	Not tested

- R17** | A17-01: print the port coupon in the intended material; measure both mounting centres and each cut; trial connectors, fasteners, actual plugs, DC nut and solder/wire access together.
- R17** | A17-02: trial actual screws in full-length 2.8 mm bosses; measure engagement and remaining pilot-depth margin. Inspect splitting/retention and all four Uno holes before approving a body change.
- R17** | A17-03/04: trial actual lid heads and pump with attached tube against matching coupons, then check seat, clearances, removal and dry boundary in the body.
- R17** | A17-05/06: measure relay hole pitch/underside; fit all four screws. Dry-build fascia, boards, wiring and planter; inspect tool access, contact/solder clearance and full removal travel.

Action ID	Actual result / evidence	Verdict / operator / date
A17-01	_____	Not tested / _____
A17-02	_____	Not tested / _____
A17-03	_____	Not tested / _____
A17-04	_____	Not tested / _____
A17-05	_____	Not tested / _____
A17-06	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action:

_____. Retest configuration/evidence: _____.

03 Power interface and relay reset acceptance

Action ID	Test	Expected acceptance	Existing status
A17-07	USB/barrel electrical interface	Exact USB pins/CC/power/data, DC polarity/voltage and source behaviour identified	Open design hold
A17-08	Relay default OFF	Coil/input/contact ratings recorded; OFF through reset and both supply sequences	Not tested

R17 | A17-07: identify USB-C part/pins and intended power/data adaptation; inspect configuration-channel provision. Measure actual DC extension polarity, mating dimensions and grip. Freeze the approved input diagram and source selection before energising it.

R17 | Record USB input, board rails and barrel input voltages/current under the agreed load. Record the intended supply arrangement and unwanted coupling tests appropriate to it. Define numerical pass limits from matching component specifications first.

R17 | A17-08: with pump disconnected, identify coil/input/contact ratings and measure contact state through board reset, open logic input and both source power-up orders. Expected contact state is OFF throughout inhibit conditions; stop on any transient closure.

R17 | Verify an accessible physical pump-power disconnect, selected fuse/wire/contact ratings and motor suppression. Do not substitute a software OFF message for measured contact state.

Action ID	Actual result / evidence	Verdict / operator / date
A17-07	_____	Not tested / _____
A17-08	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action:

_____. Retest configuration/evidence: _____.

04 LCD, controls and dose measurements

Action ID	Test	Expected acceptance	Existing status
A17-09	LCD and controls	Display stable, buttons/lamps verified, dose return and full 60 s timeout observed	Partial serial record only
A17-10	Dose and volume	Actual contact/pump durations and volume measured at selected endpoints	Not measured

R17	A17-09: with pump supply OFF, verify each switch and lamp; stable LCD status 0 and visible pages; STOP/release lockout; 1-4 s indicated endpoints; safe soil-index/fault test inputs; no automatic request from moisture.
R17	Observe and record the actual LCD backlight after 60 s idle, button/knob wake and 60 s after completed/cancelled requests. Confirm immediate soil Status return at request end. Retain video/timed observation and serial as separate evidence.
R17	A17-10: after power/reset protections pass, run a supervised measured pump trial using the approved supply and installed tube/lift. Measure contact ON duration and delivered volume at 1.0/4.0 s settings and intermediate points required by the intended use.
R17	Timing and volume acceptance tolerances: Not yet defined. Record instrumentation resolution, repetitions, individual readings, mean/range and failures. Do not mark a test Pass without the pre-set numerical criteria.

Action ID	Actual result / evidence	Verdict / operator / date
A17-09	_____	Not tested / _____
A17-10	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action: _____.

Retest configuration/evidence: _____.

05 Leak/fill, print configuration and CAD exception

Action ID	Test	Expected acceptance	Existing status
A17-11	Leak/fill and wet route	Leak, drain-back, tilt/splash evidence; fill limit below unsealed openings recorded	Not tested
A17-12	Material/print configuration	Material, slicer layers, orientation, local fit/strength recorded	Unknown
A17-13	CAD shutdown anomaly	Clean CAD process log or isolated dependency defect with retained evidence	Open; build exits 1

- R17** | A17-11: define leak-test duration, fill candidate, containment and acceptance first. Record actual water level, drain-back headspace, soil/drain condition, planter movement, tilt/splash exposure and dry-tower inspection. Fill mark acceptance needs physical evidence.
- R17** | A17-12: record material and printing process, orientation/supports, layer inspection and measured critical holes. Representative screw and connector coupons must match the material/process of the intended body.
- R17** | A17-13: retain the build output and exit code. Current R17 builder prints PASS but exits 1 during shutdown; independent STL checker exits 0. Closing this exception needs a clean process or isolated dependency-defect evidence; it never closes physical fit.

Action ID	Actual result / evidence	Verdict / operator / date
A17-11	_____	Not tested / _____
A17-12	_____	Not tested / _____
A17-13	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action:
 _____. Retest configuration/evidence: _____.

06 Close-out and disposition

Action	Result	Evidence / owner / target
A17-01	Not tested / Open	_____
A17-02	Not tested / Open	_____
A17-03	Not tested / Open	_____
A17-04	Not tested / Open	_____
A17-05	Not tested / Open	_____
A17-06	Not tested / Open	_____
A17-07	Not tested / Open	_____
A17-08	Not tested / Open	_____
A17-09	Not tested / Open	_____
A17-10	Not tested / Open	_____
A17-11	Not tested / Open	_____
A17-12	Not tested / Open	_____
A17-13	Not tested / Open	_____

R17

Closed actions: None pre-populated. Open defects/deviations: _____.
Reviewer/name/date: _____. Accepted configuration and limits: _____.

CAUTION
Current disposition remains prototype review with physical acceptance open. A signed worksheet closes only the identified test article and recorded criteria; it does not infer an ingress rating, certification or unattended-operation acceptance.